Lecture_11_Curves_and_Basic_Animation

CSTE 3201: Computer Graphics

Lecture 11: Curves and Basic Animation

> **Combined focused lecture from selected parts of Lecture 11 and Lecture 12**
> Main focus: **Bezier curves, splines, and basic animation ideas.**

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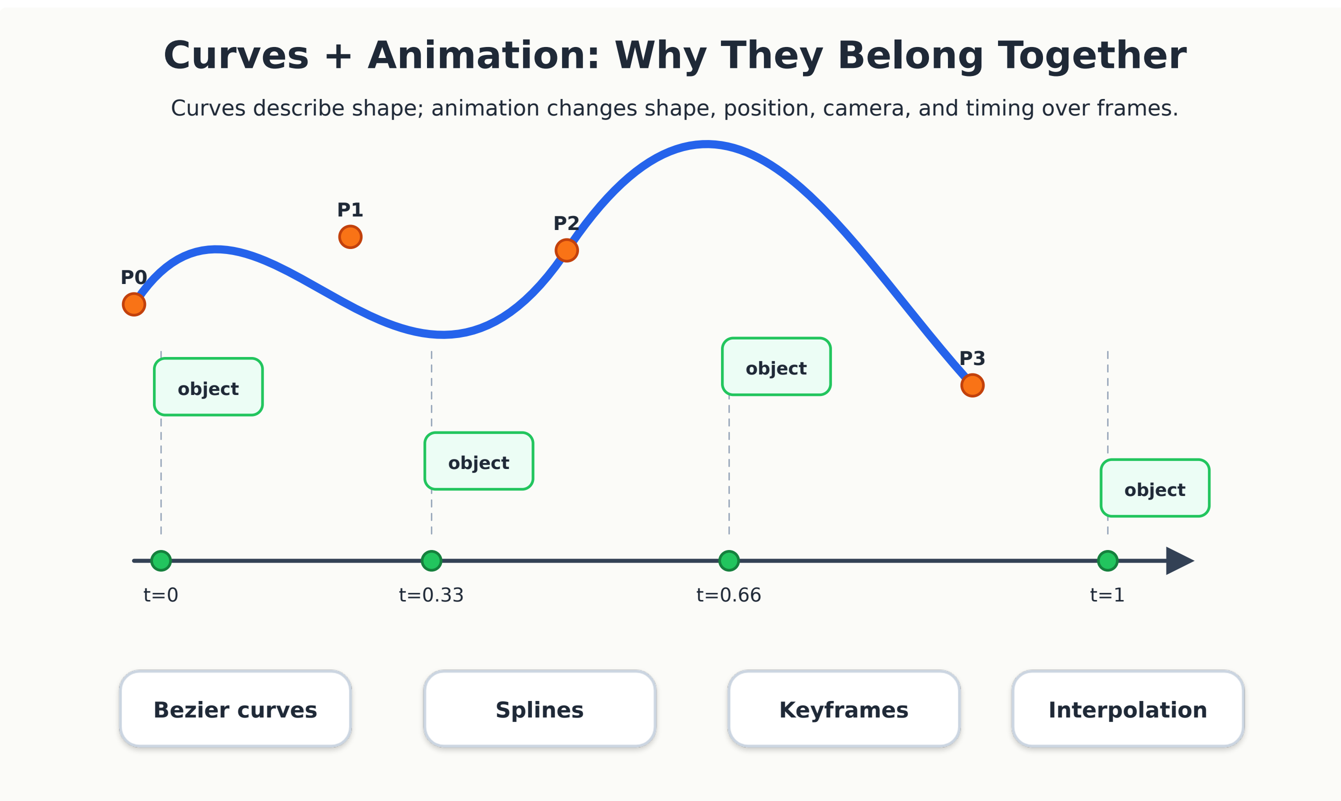
1. Learning Objectives

1. Explain why curves are important in computer graphics.
2. Represent a curve parametrically using a parameter (t).
3. Describe Bezier curves using control points.
4. Explain the role of Bernstein blending functions.
5. Use the de Casteljau algorithm conceptually.
6. Explain why splines are needed for long and complex curves.
7. Distinguish between (C^0), (C^1), and (C^2) continuity at curve joins.
8. Explain the basic idea of key-frame animation.
9. Understand interpolation, tweening, easing, and motion along a path.



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2. Why Curves Matter in Computer Graphics

Many objects in the real world are not made of straight lines only. Examples include:

- roads,
- rivers,
- product outlines,
- character bodies,
- camera paths,
- handwriting,

- fonts,
- vehicle motion paths,
- animation trajectories.

In early graphics, objects were often approximated by many short straight line segments. This works, but it can be inefficient and hard to edit. A curve representation gives a more compact and controllable way to describe smooth shapes.

For example, instead of storing hundreds of small line segments for a smooth road, we can store a small number of control points and generate the smooth road from them.

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3. Parametric Curves

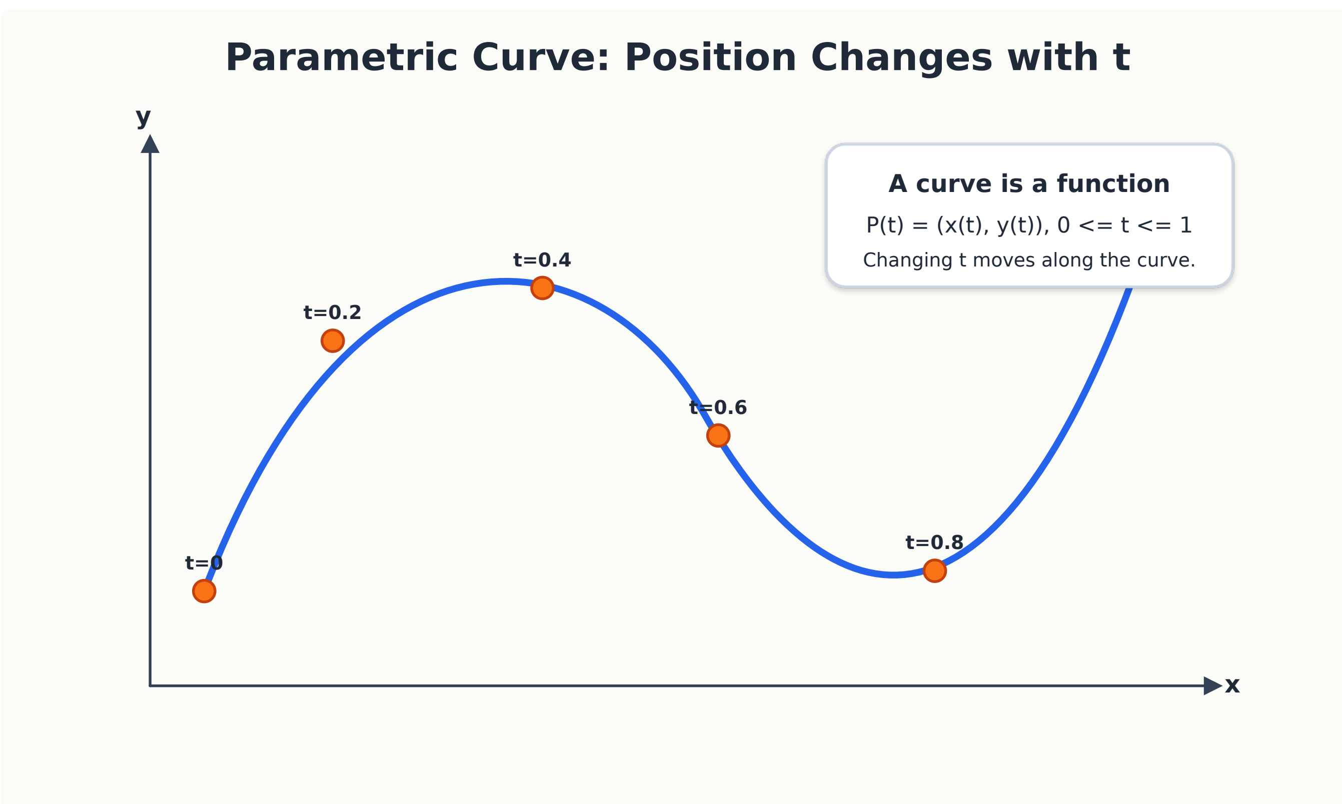
A curve can be described as a function of a parameter (t). Usually, (t) varies from 0 to 1.

$$P(t) = (x(t), y(t)), \quad 0 \le t \le 1$$

For 3D curves:

$$P(t) = (x(t), y(t), z(t))$$

Here, t does not necessarily mean time, but in animation it is often connected to time.



Intuition

Think of t as a slider:

- when $t = 0$, we are at the beginning of the curve;
- when $t = 1$, we are at the end of the curve;
- intermediate values of t give intermediate points on the curve.

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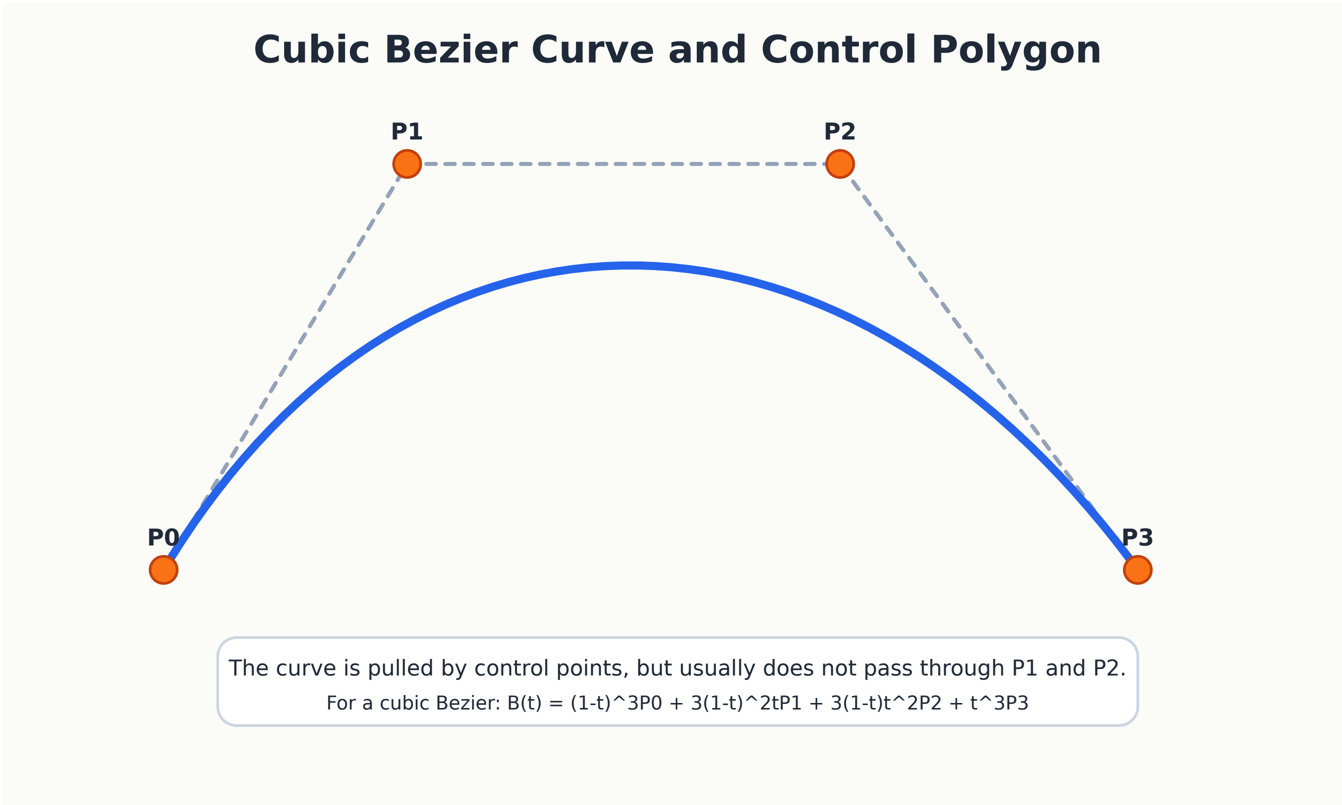
4. Bezier Curves

Bezier curves are among the most widely used curves in computer graphics, vector drawing, font design, CAD, and animation.

A Bezier curve is controlled by a set of **control points**. The curve does not always pass through all control points. Instead, the control points pull the curve toward themselves.



4.1 Linear Bezier Curve

With two control points P_0 and P_1:

$$B(t) = (1-t)P_0 + tP_1$$

This is just a straight line segment from (P_0) to (P_1).

4.2 Quadratic Bezier Curve

With three control points (P_0, P_1, P_2):

$$B(t) = (1-t)^2 P_0 + 2(1-t)tP_1 + t^2 P_2$$

The curve starts at (P_0), ends at (P_2), and is influenced by (P_1).

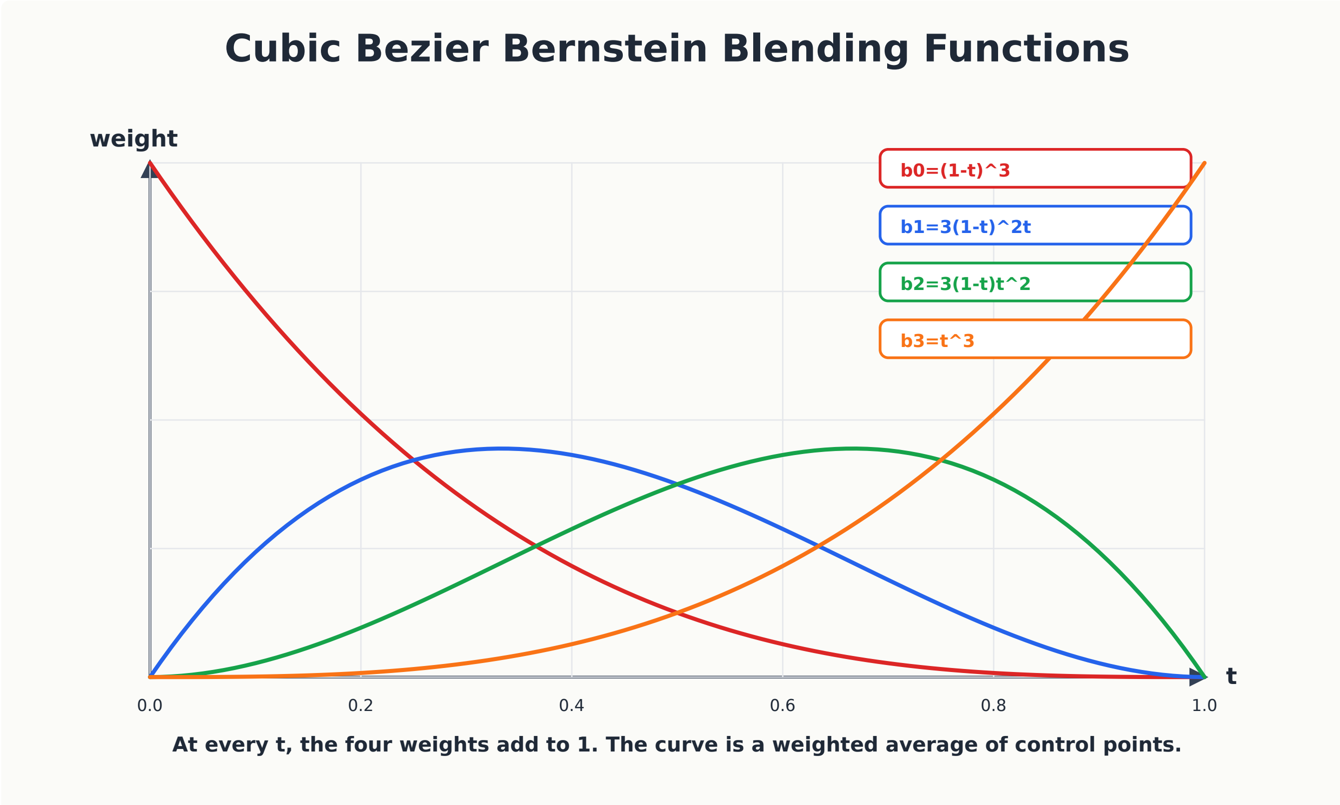
4.3 Cubic Bezier Curve

The cubic Bezier curve is the most common form in graphics software.

With four control points (P_0, P_1, P_2, P_3):

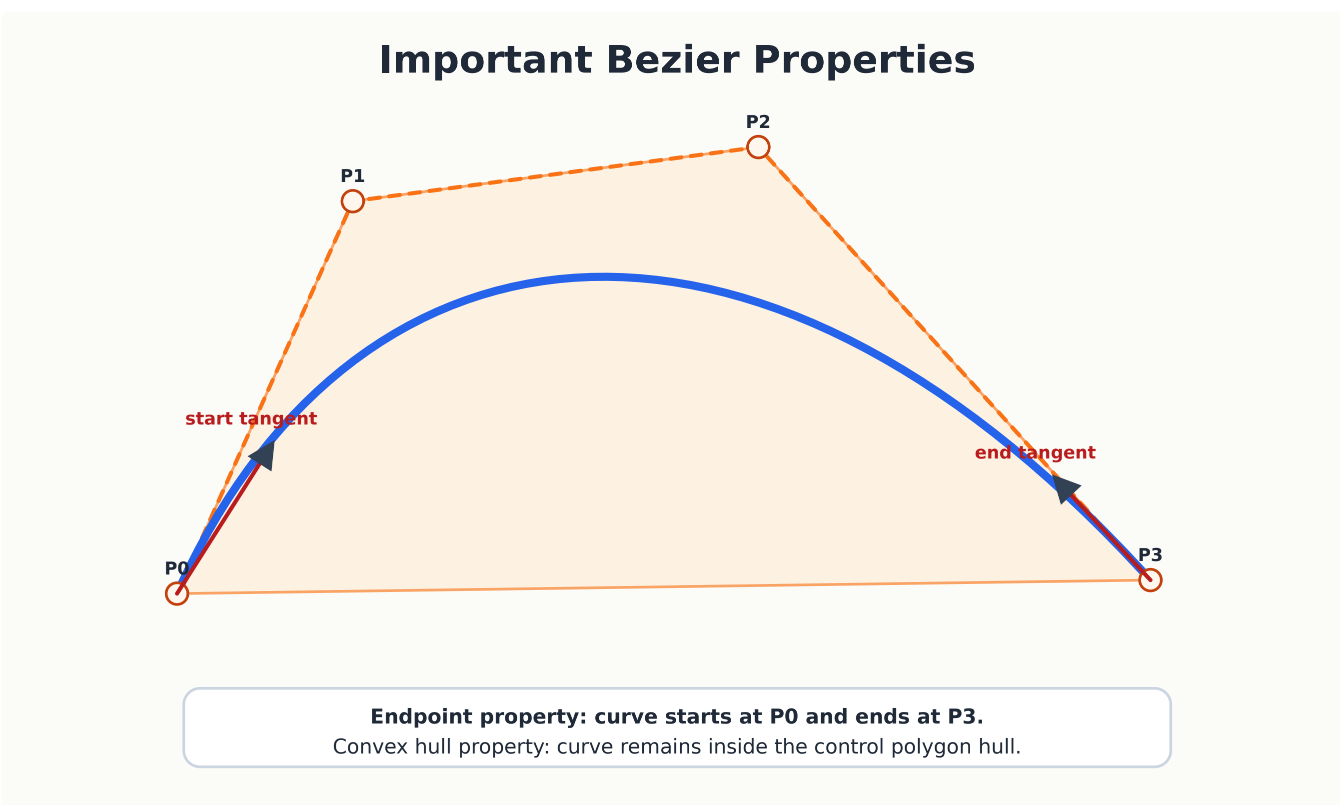
$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 tP_1 + 3(1-t)t^2 P_2 + t^3 P_3$$

The four coefficients are called **Bernstein blending functions**.



Important Properties

1. The curve starts at P_0.
2. The curve ends at P_3.
3. The curve is pulled by P_1 and P_2.
4. The curve lies inside the convex hull of its control points.
5. The tangent at the start follows the direction P_0P_1.
6. The tangent at the end follows the direction P_2P_3.



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5. Worked Example: Evaluate a Cubic Bezier Curve

Suppose a cubic Bezier curve has control points:

$$P_0 = (0,0), \quad P_1 = (1,2), \quad P_2 = (3,2), \quad P_3 = (4,0)$$

Find $(B(0.5))$.

Using:

$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3$$

At (t=0.5):

$$(1-t)^3 = 0.125$$

$$3(1-t)^2 t = 0.375$$

$$3(1-t)t^2 = 0.375$$

$$t^3 = 0.125$$

So:

$$B(0.5) = 0.125P_0 + 0.375P_1 + 0.375P_2 + 0.125P_3$$

For x-coordinate:

$$x = 0.125(0) + 0.375(1) + 0.375(3) + 0.125(4) = 2$$

For y-coordinate:

$$y = 0.125(0) + 0.375(2) + 0.375(2) + 0.125(0) = 1.5$$

Therefore:

$$B(0.5) = (2, 1.5)$$

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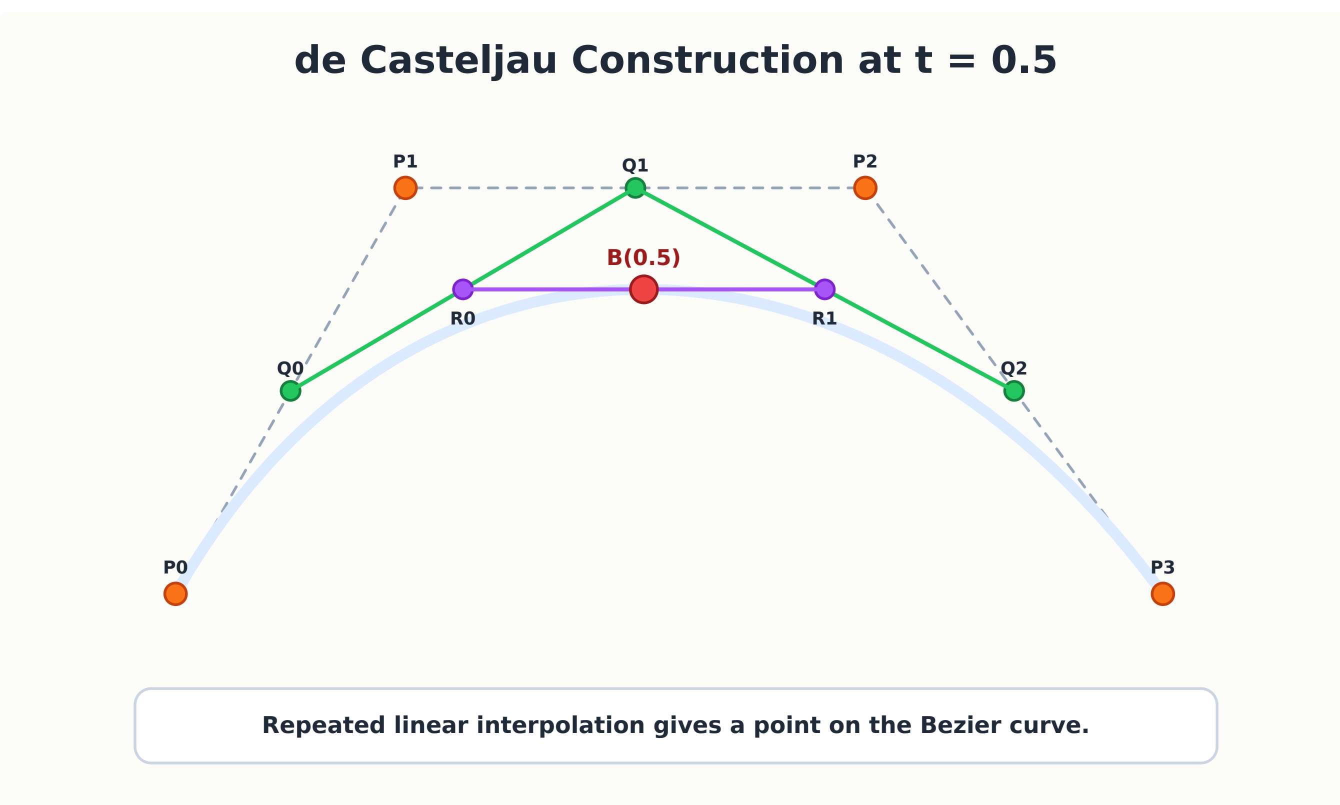
6. de Casteljau Algorithm

de Casteljau's algorithm is a geometric way to evaluate a Bezier curve.

Instead of directly using the polynomial formula, it repeatedly performs linear interpolation between points.



For a Cubic Bezier Curve

Given (P_0, P_1, P_2, P_3):

First level:

$$Q_0 = (1-t)P_0 + tP_1$$

$$Q_1 = (1-t)P_1 + tP_2$$

$$Q_2 = (1-t)P_2 + tP_3$$

Second level:

$$R_0 = (1-t)Q_0 + tQ_1$$

$$R_1 = (1-t)Q_1 + tQ_2$$

Final point:

$$B(t) = (1-t)R_0 + tR_1$$

Pseudocode

```
function deCasteljau(points, t):
    temp = copy(points)
    n = number of points

    for level = 1 to n - 1:
        for i = 0 to n - level - 1:
            temp[i] = (1 - t) * temp[i] + t * temp[i + 1]

    return temp[0]
```

Why It Is Important

The de Casteljau algorithm is:

- geometrically intuitive,
- numerically stable,
- useful for curve subdivision,
- useful for rendering and editing Bezier curves.

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7. Why Splines Are Needed

A single Bezier curve works well for simple shapes. But many real shapes are too long or too complex for one Bezier segment.

For example:

- a road map,
- a character outline,
- a handwriting stroke,
- an animation path,
- a product surface outline.

Using one high-degree Bezier curve would be difficult to control. Moving one control point could affect a very large portion of the curve.

A **spline** solves this problem by joining several curve segments together.

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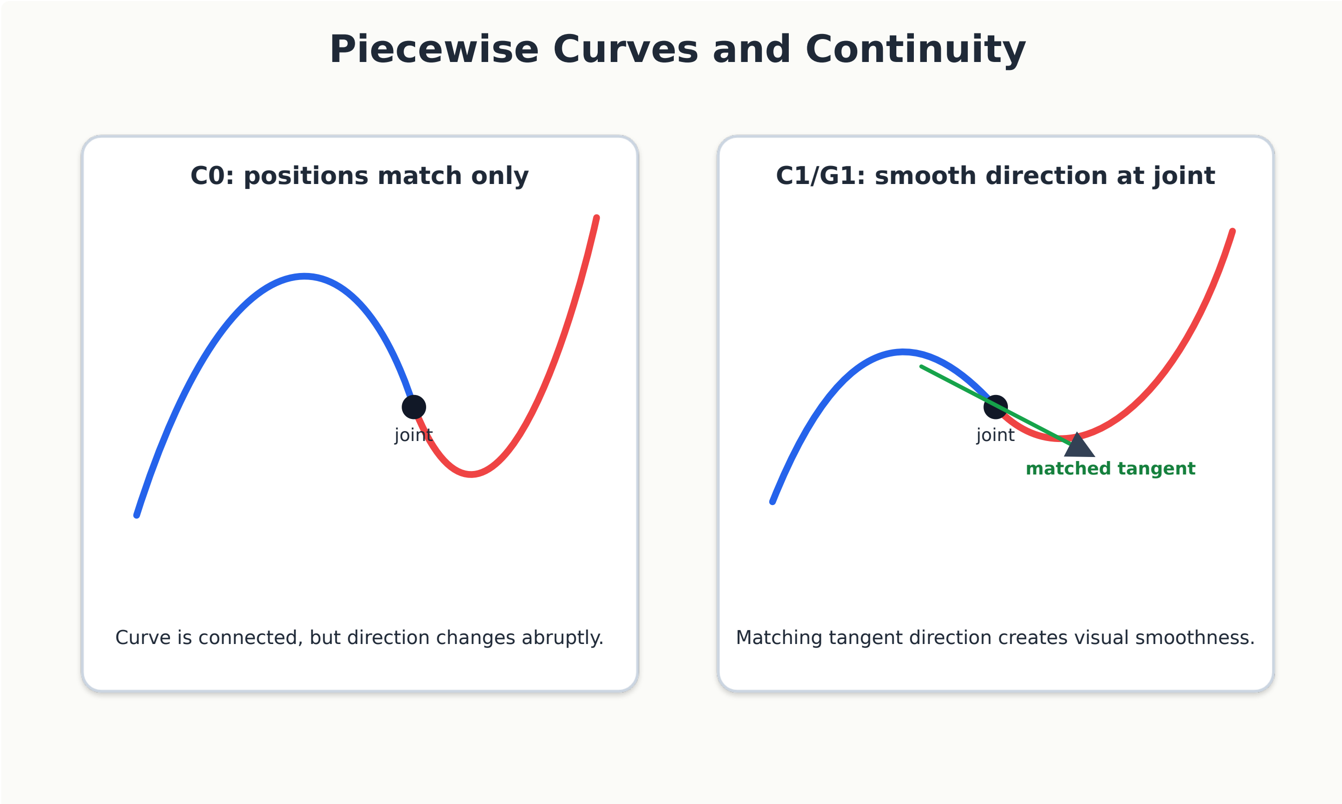
8. Continuity in Piecewise Curves

When two curve segments are joined, we need to consider how smooth the join is.



8.1 C^0 Continuity

The two curve segments meet at the same point.

This means the curve is connected, but it may have a sharp corner.

8.2 C^1 Continuity

The two curve segments meet at the same point and have the same first derivative at the join.

This means the direction of the curve is smooth at the join.

8.3 C^2 Continuity

The two curve segments have matching position, first derivative, and second derivative at the join.

This gives smoother curvature and is important for high-quality paths and surfaces.

Geometric Continuity

Sometimes we use G^1 continuity instead of C^1.
G^1 means the tangent direction is the same, even if the speed along the curve is different.

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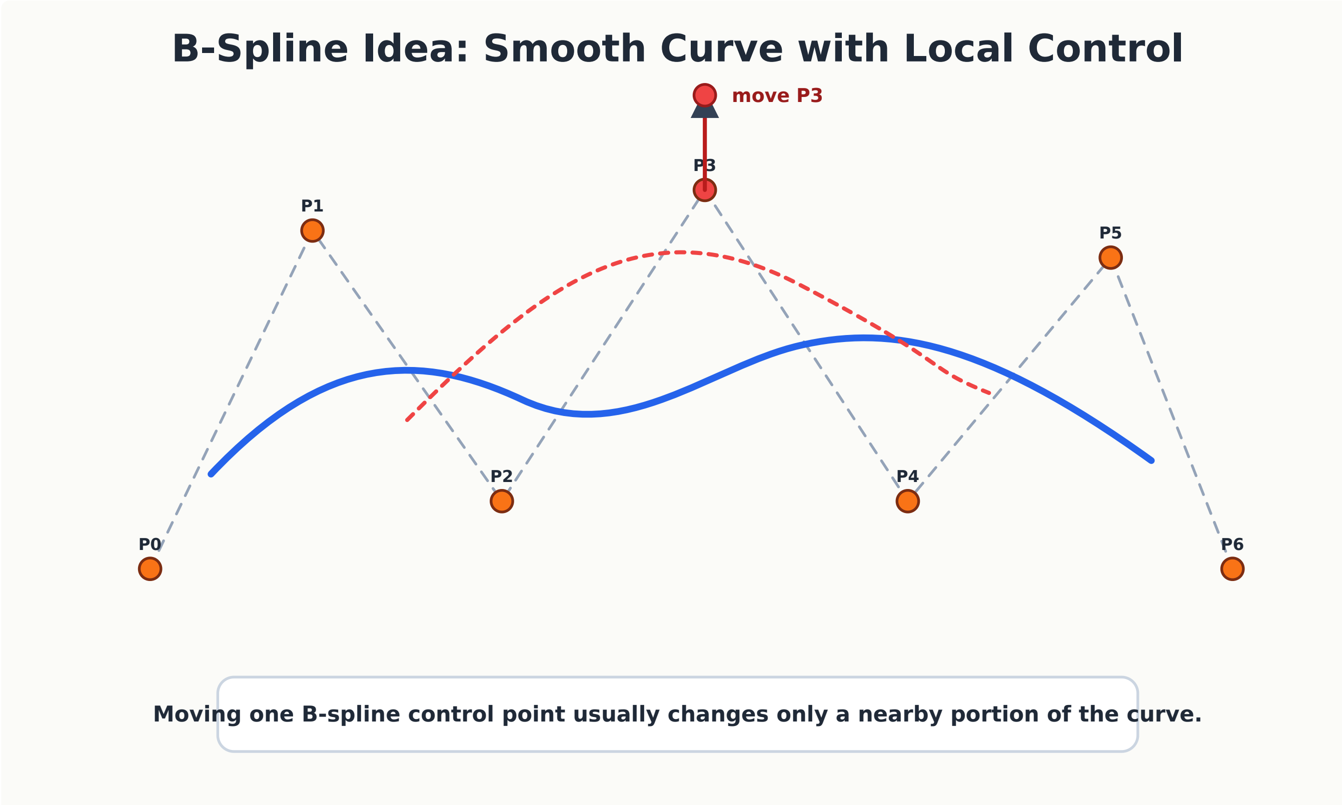
9. B-Spline Idea

A B-spline is a smooth curve controlled by several control points.

The most important idea for this course is **local control**.

If one control point is moved, only a nearby portion of the curve changes. The entire curve does not change dramatically.



Bezier vs B-Spline

Feature	Bezier Curve	B-Spline Curve
Control	Global effect for one segment	More local control
Shape	Good for short smooth segments	Good for long smooth curves
Common use	Vector graphics, fonts, paths	CAD, modeling, animation paths
Editing	Simple for small curves	Better for complex curves

For CT-level understanding, remember:

> A spline is a piecewise smooth curve, and B-splines are useful because they provide smoothness and local control.

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10. Basic Idea of Animation

Animation is the illusion of motion created by showing a sequence of images or frames quickly.

A frame is one still image. If frames are displayed rapidly, the human visual system perceives motion.

Common frame rates:

- 24 fps: cinema,
- 30 fps: many videos,
- 60 fps: smooth games and displays.

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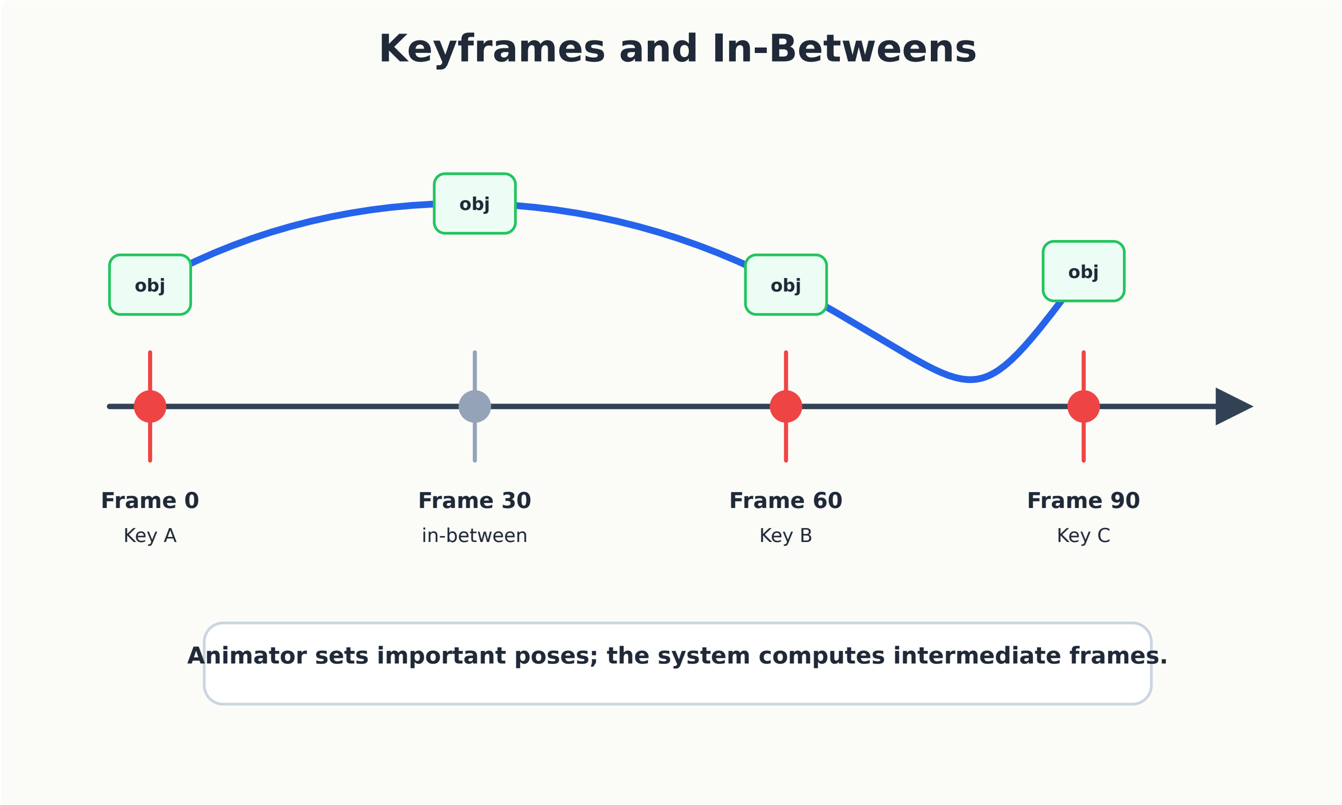
11. Key-Frame Animation

In key-frame animation, the animator specifies important frames called **keyframes**. The computer then generates intermediate frames.



Example:

- Frame 0: ball is on the left.
- Frame 30: ball is at the top.
- Frame 60: ball is on the right.

The frames between them are generated by interpolation.

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12. Interpolation and Tweening

Tweening means generating in-between frames between keyframes.

Suppose an object moves from position P_0 to P_1.

A simple linear interpolation is:

$$P(t) = (1 - t)P_0 + tP_1$$

where $0 \leq t \leq 1$.

This is the same basic idea used in linear Bezier curves.

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13. Easing

Linear interpolation gives constant speed. But real motion often does not look like that.

An object may:

- start slowly and speed up,
- start fast and slow down,
- bounce,
- overshoot,
- oscillate.

This is where **easing functions** are used.

Interpolation and Easing

Easing changes timing, not the start and end positions.

Examples

Easing type	Behavior
Linear	Constant speed
Ease-in	Starts slowly, then speeds up
Ease-out	Starts fast, then slows down
Ease-in-out	Starts slow, speeds up, then slows down

Easing changes the timing of motion, not necessarily the start or end position.

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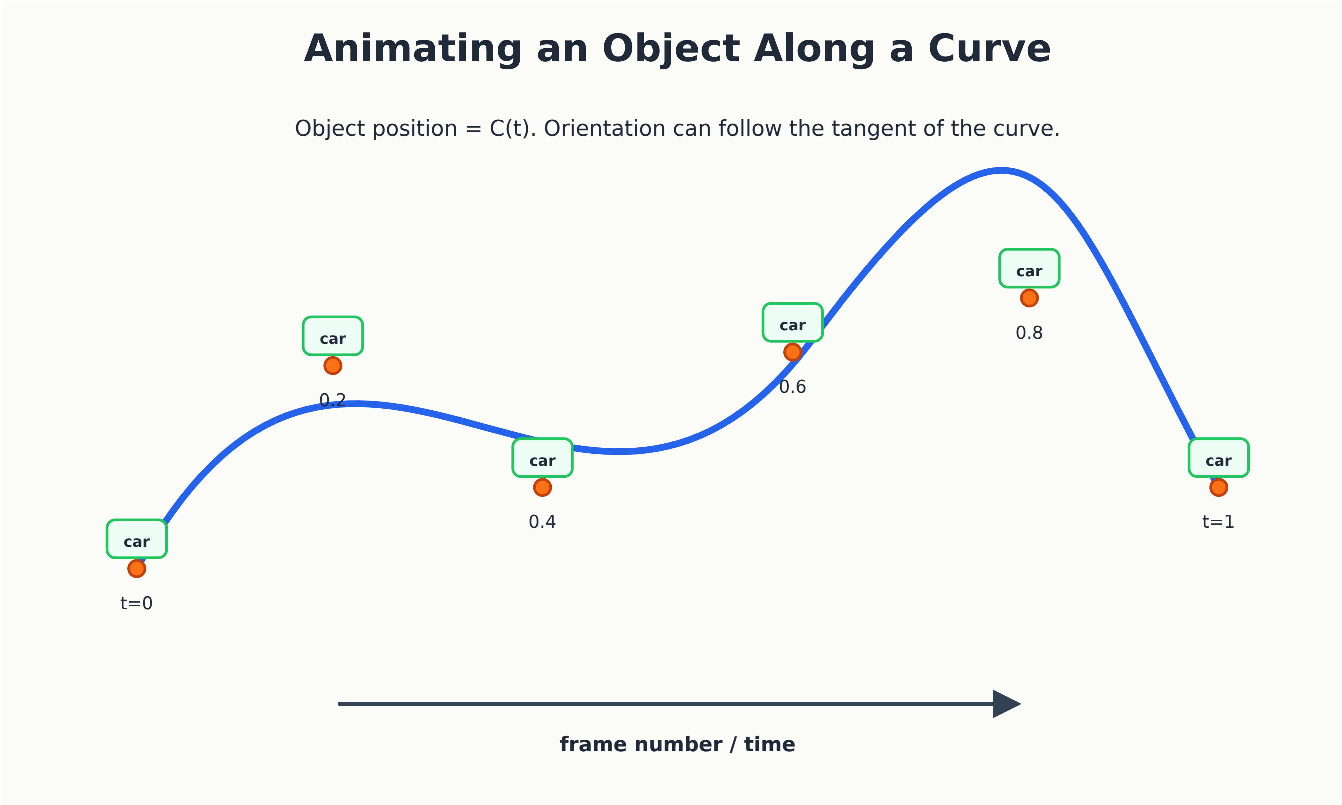
14. Motion Along a Curve

Curves are useful not only for drawing shapes, but also for defining motion paths.

If $C(t)$ is a curve, then an animated object can be placed at:

$$P = C(t)$$

for different values of t over time.



For example:

- a car following a road,
- a camera moving through a scene,
- a flying object following a smooth path,
- a UI element sliding smoothly along a curve.

The object's orientation can also follow the tangent direction of the curve.

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15. Animation Using Transformations

In earlier lectures, we studied transformations:

- translation,
- rotation,
- scaling,
- reflection,
- shear.

In animation, these transformations change over time.

Example:

$$T(t) = \text{translation by } (x(t), y(t))$$

If $x(t)$ and $y(t)$ change every frame, the object appears to move.

Similarly:

- changing rotation angle over time makes an object spin;
- changing scale over time makes an object grow or shrink;
- changing color or opacity over time creates visual effects.

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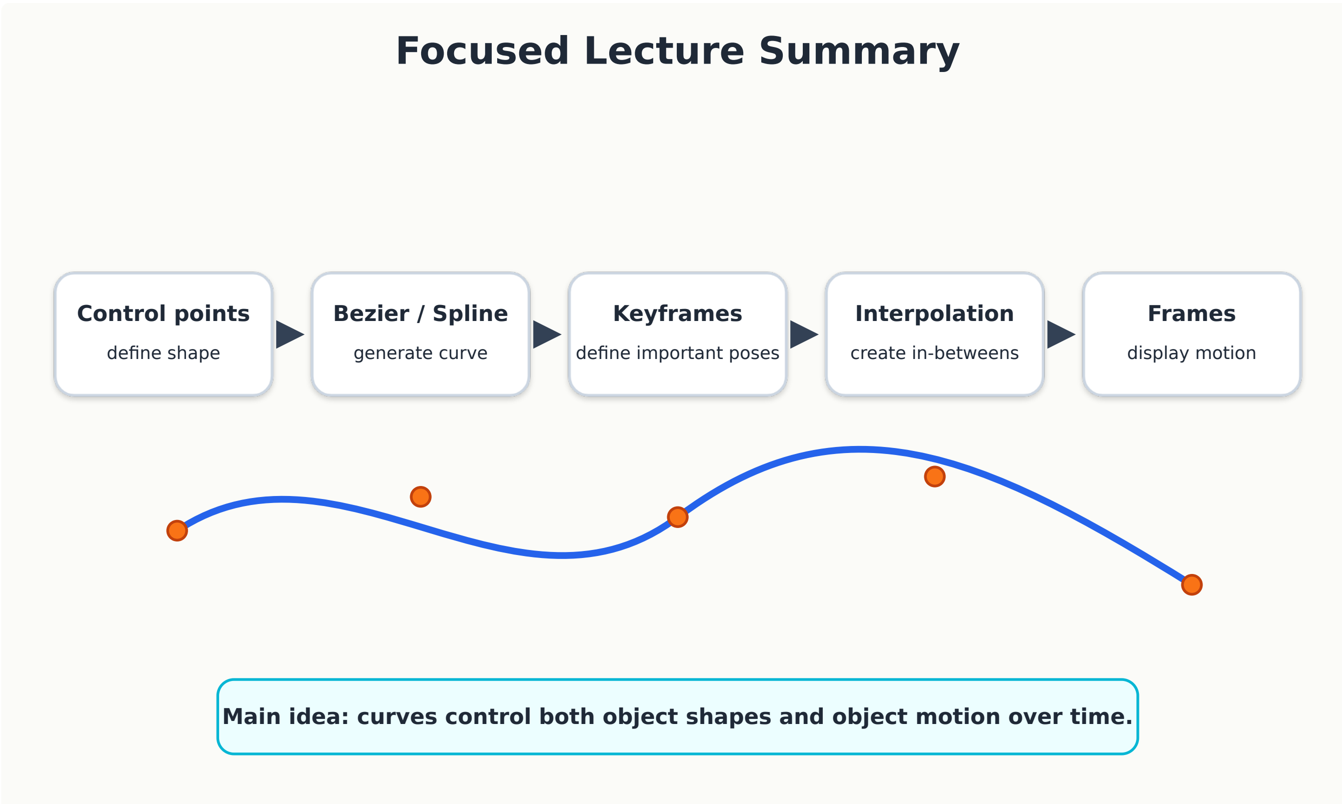
16. Simple Animation Pipeline

A simplified animation pipeline is:

1. Design objects or models.
2. Define keyframes or motion paths.
3. Interpolate positions, rotations, scales, or other properties.
4. Render each frame.
5. Display frames in sequence.



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17. Mini Example: Linear Motion Between Two Keyframes

An object moves from $P_0 = (2, 3)$ to $P_1 = (10, 7)$.

Find the position at $t = 0.25$.

$$P(t) = (1 - t)P_0 + tP_1$$

$$P(0.25) = 0.75(2, 3) + 0.25(10, 7)$$

$$P(0.25) = (1.5, 2.25) + (2.5, 1.75)$$

$$P(0.25) = (4, 4)$$

So, at 25% of the animation time, the object is at:

$(4, 4)$

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18. Common Student Mistakes

Mistake 1: Thinking Bezier curves pass through every control point

A Bezier curve passes through the first and last control points, but generally does not pass through the middle control points.

Mistake 2: Confusing control polygon with the curve

The control polygon is a guide. It is not the final curve.

Mistake 3: Thinking all smooth curves are Bezier curves

Bezier curves are one type of curve. Splines and B-splines are also important.

Mistake 4: Thinking keyframes are all frames

Keyframes are important frames set by the animator. In-between frames are generated by interpolation.

Mistake 5: Thinking easing changes only the path

Easing mainly changes the rate of progress over time. The path may remain the same.

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19. Applications

Bezier Curves

- vector graphics,
- font outlines,
- logo design,
- drawing tools,
- SVG paths,
- UI design,
- simple animation paths.

Splines and B-Splines

- CAD modeling,
- car body design,
- character modeling,
- road and railway design,
- smooth camera paths,
- 3D surface design.

Animation

- cartoons,
- games,
- simulations,
- user interface transitions,
- motion graphics,
- virtual reality,
- robotics visualization.

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20. Short Trivia

- Bezier curves are named after **Pierre Bezier**, who used them in automobile design at Renault.
- Similar ideas were also developed by **Paul de Casteljau** at Citroen.
- Modern vector graphics formats such as SVG use Bezier curves heavily.
- Fonts on computers often use curve-based outlines rather than storing every pixel of every letter.
- In animation software, the graph editor often shows curves that control motion timing.

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21. Quick Check Questions

Q1

A cubic Bezier curve has how many control points?

Answer: Four.

Q2

Does a cubic Bezier curve generally pass through P_1 and P_2?

Answer: No. These points influence the curve but are not usually on the curve.

Q3

What is the main advantage of a B-spline over a single high-degree Bezier curve?

Answer: Better local control and smoother handling of complex curves.

Q4

What does key-frame animation require from the animator?

Answer: Important poses or states at selected frames.

Q5

What does interpolation compute in animation?

Answer: Intermediate values between keyframes.

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22. Practice MCQs

MCQ 1

Which statement about a cubic Bezier curve is most accurate?

A. It always passes through all four control points.
B. It starts at the first control point and ends at the last control point.
C. It cannot be used for animation paths.
D. It is only useful for raster images.

Correct answer: B

MCQ 2

What is the main purpose of the middle control points in a cubic Bezier curve?

A. They determine the image resolution.
B. They pull or shape the curve.
C. They define the color of the curve.
D. They determine the frame rate of animation.

Correct answer: B

MCQ 3

Why are splines useful for long complex curves?

A. They are always straight lines.
B. They join smaller curve segments smoothly.
C. They remove the need for control points.
D. They work only for bitmap images.

Correct answer: B

MCQ 4

In animation, what is an in-between frame?

A. A frame manually drawn before every keyframe.
B. A frame computed between two keyframes.
C. A frame that contains no object.
D. A frame used only for clipping.

Correct answer: B

MCQ 5

Which technique makes motion start slowly and then speed up?

A. Clipping
B. Ease-in
C. Z-buffering
D. Boundary fill

Correct answer: B

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23. Exam-Oriented Summary

For exams and CTs, focus on these ideas:

1. A parametric curve uses a parameter (t) to generate points.
2. A Bezier curve is controlled by control points.
3. A cubic Bezier curve uses four control points.
4. The first and last control points lie on the curve.
5. Middle control points influence the curve shape.
6. de Casteljau algorithm uses repeated linear interpolation.
7. A spline is a smooth piecewise curve.
8. B-splines are useful because of local control.
9. Animation is produced by displaying frames in sequence.
10. Keyframes are important frames; in-betweens are computed by interpolation.
11. Easing changes how motion progresses over time.
12. Curves can be used as motion paths.

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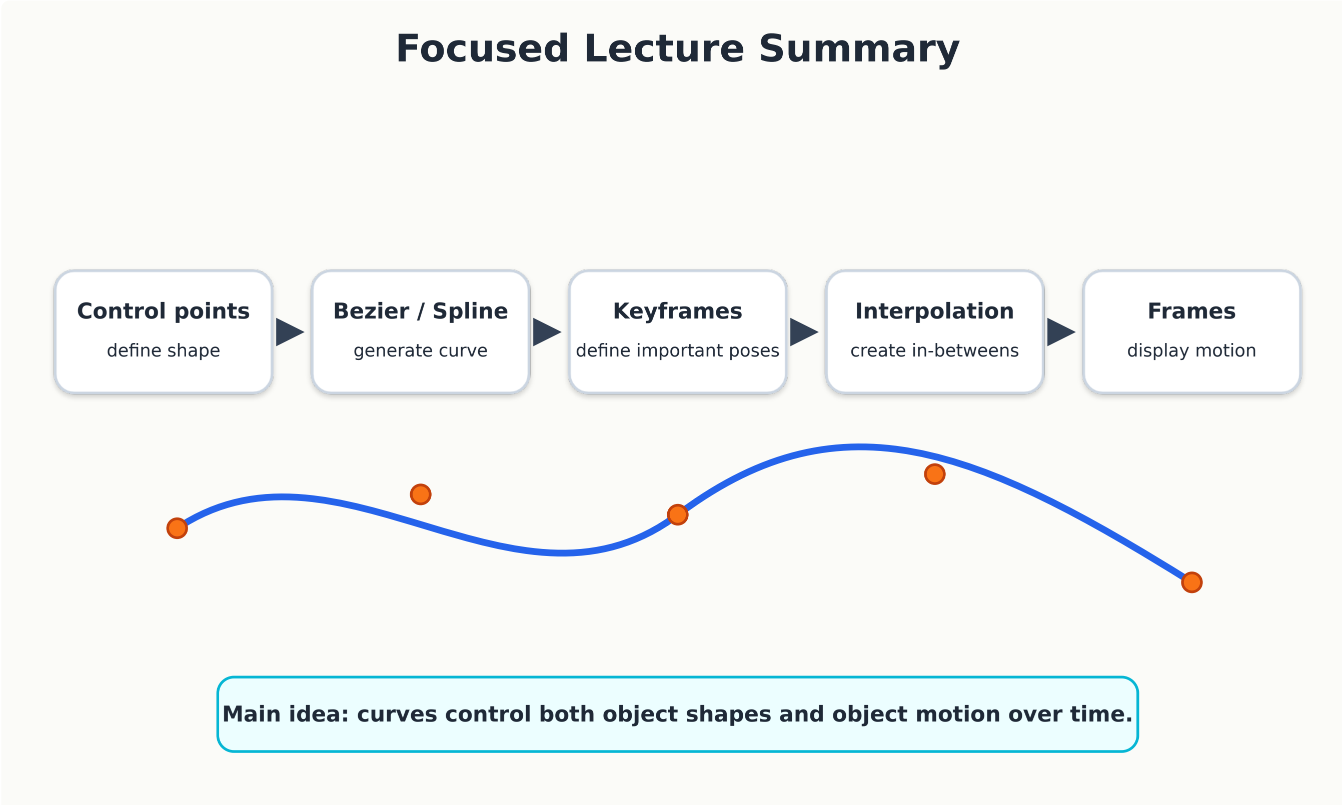
24. One-Slide Final Summary



A compact way to remember this lecture:

> **Curves describe smooth shapes and paths. Animation changes objects over time. Interpolation connects both ideas.**